rag -- 2026-11-12

5 new paper(s) · topic: Retrieval-augmented generation: retrieval quality, grounding and faithfulness, evaluation benchmarks, and RAG system design.

Worth reading in full

[P1] CLADE: Category-Level Answering from Deficient Evidence in Retrieval-Augmented Language Models -- P1 provides a detailed analysis of how retrieval granularity directly harms faithfulness and proposes concrete mitigation attempts, making it the most informative for anyone studying retrieval-augmented generation's grounding and evaluation challenges.

Themes

Several papers examine how the granularity and relevance of retrieved evidence shape downstream generation. CLADE shows that when retrieval returns only a parent-category ("family-only") passage, models dramatically lower abstention and make roughly half wrong guesses, and that simple confidence filtering or scaling model size only modestly helps [P1]. AOP-Smart-Finder uses a retrieval-augmented generation engine that accesses the official AOP-Wiki knowledge base and combines high-throughput LLM screening with manual curation and weight-of-evidence evaluation, achieving 100 % accuracy on a random check of 100 mapped key events [P2]. ROAM demonstrates a retrieval-augmented optimization loop that fetches a compact set of points-of-interest from a semantic embedding index and then tightly integrates spatial constraints, showing that retrieval can be tightly coupled to downstream combinatorial planning to improve feasibility and route efficiency [P3]. TIGA treats taxonomy induction as an iterative decision process using tool-grounded agents that perform actions such as add_entity, remove_entity, view_subtree, and update_policy, improving recall and edge-level F1 while incurring higher computational cost and showing variable precision compared with baselines [P4]. Definitions, Label Names, or Demonstrations? audits how retrieval-augmented prompting (Global Example-RAG) boosts annotation accuracy without increasing adherence to supplied definitions, revealing that retrieval can improve performance metrics while the model still relies on prior label mappings rather than true grounding to the provided codebook [P5]. Across these works, a recurring evaluation gap emerges: most studies report aggregate metrics (precision, recall, coverage, F1) but lack systematic human-centered or factuality-focused benchmarks that directly measure grounding fidelity, especially under ambiguous or coarse evidence conditions.

5 theme sentence(s) checked against the paper summaries; 2 corrected, 0 removed.

At a glance

[P1] CLADE: Category-Level Answering from Deficient Evidence in Retrieval-Augmented Language Models · relevance 0.28
Retrieval-augmented models often guess specific answers when only a coarse family name is present, a behavior called Clade that lowers abstention but yields many wrong guesses and resists simple fixes.

[P2] AOP-Smart-Finder: A High-Throughput Human-AI Hybrid Framework for AOP Discovery and Evidence Mapping from Large-Scale Literature · relevance 0.27
AOP-Smart-Finder is a high-throughput human-AI hybrid system that uses large language models and official AOP-Wiki knowledge to screen tens of thousands of papers, extract thousands of biological events and key-event relationships, and build curated mechanistic networks for toxicity and disease studies.

[P3] ROAM: Retrieval-Augmented Optimization for Feasible Multi-Day Itinerary Planning · relevance 0.25
ROAM dramatically increases the number of feasible attractions in multi-day itineraries while cutting travel dispersion by using retrieval-augmented optimization [1].

[P4] Controlled Knowledge Engineering for Taxonomy Induction with Tool-Grounded LLM Agents and Runtime-Adaptive Policies · relevance 0.25
TIGA is a tool-grounded LLM agent framework that improves taxonomy induction's recall and overall F1 by using explicit state, constrained actions, memory, and adaptive policies, but it incurs substantially higher computational cost and does not uniformly beat baselines on precision.

[P5] Definitions, Label Names, or Demonstrations? Auditing What Governs LLM Annotation · relevance 0.25
The paper audits whether LLMs annotate texts by following supplied codebook definitions or by preserving original label mappings, finding that models usually stick to the original mapping and that retrieval-augmented prompting boosts accuracy without improving definition adherence.

[P1] CLADE: Category-Level Answering from Deficient Evidence in Retrieval-Augmented Language Models

S M Rayeed, Charles V. Stewart, Yao Ma · arXiv 2611.14355v1 · https://arxiv.org/abs/2611.14355v1 · relevance 0.28

TL;DR: Retrieval-augmented models often guess specific answers when only a coarse family name is present, a behavior called Clade that lowers abstention but yields many wrong guesses and resists simple fixes.

Problem
Retrieval-augmented language models receive passages that mention only a broad category (e.g., a disease family) instead of the exact answer required by the question, creating "resolvable insufficiency" where the text alone cannot pinpoint the fine-grained answer but points to a short list of candidates in an ontology [1]. This situation leads models to answer rather than abstain, a phenomenon the authors name Clade [1].

Method
The authors categorize retrieved evidence into three granularity levels--named (exact name), family-only (parent category within two levels), and silent (neither)--and evaluate models on biomedical drug-disease facts from PrimeKG and ecological species interactions from GloBI [2][3]. They conduct counterfactual edits that replace exact disease names with family names or delete the sentence, and they test three mitigation attempts: confidence-based filtering, scaling model size, and explicit abstention instructions [3]. They also assess whether an ontology can recover exact answers from the family name by string matching and by letting the model select the family [1][4].

Key results
- Family-only passages reduce abstention by 13.8 percentage points relative to silent passages on PrimeKG (and by 14.3 points on GloBI) [3].
- The resulting guesses are wrong about half the time (~50 % error rate) [1].
- Counterfactual edits show that keeping a family name (instead of deleting the sentence) cuts abstention by 17.4 points and raises wrong answers by 5.1 points; replacing an exact name with a family name lowers correct answers by 21.0 points and raises wrong answers by 9.5 points [3].
- Confidence filtering does not separate correct from wrong answers on family-only pages (AUROC 0.59-0.72 vs. 0.73-0.87 on named pages) [3].
- Scaling to larger models improves precision on family-only pages only up to a point (38.1 % -> 59.4 % at 7 B, then stalls at ~58 % for 14 B) [3].
Instruction to abstain raises abstention by about 15 points on family-only pages, also raises abstention on silent and named pages, costs about 6.3 points of correct answers on named pages, and precision on family-only pages stays between roughly 48 % and 65 %, far below the 90 %+ precision on named pages [3].
- Ontology recovery: given the correct family, the MONDO descendant list contains the true answer in 98.5 % of cases, but exact string matching finds the family on the page only 19.8 % of the time; letting the model choose the family raises this to 30.0 %, and end-to-end recovery improves from 10.0 % to 26.8 % [1].

- Closed-book recall on Qwen2.5-7B shows correct answers for 39.6 % of family-only facts, 32.0 % of silent facts, and 65.7 % of named facts [2].

Limitations
The study is limited to two domains (biomedical PrimeKG and ecological GloBI) and performs ontology recovery only on PrimeKG [4]. Experiments use multiple-choice questions with an explicit decline option, which may not reflect free-text refusals, although a free-text check shows similar patterns [4]. Only open models up to 14 B parameters are evaluated, and page edits cover only named facts whose answer has a parent category [4]. Page-built lists rely on exact string matching, and there are few family-only contraindications (106 questions) [4].

Sources

[1] arXiv 2611.14355v1.pdf, arXiv abstract

[2] arXiv 2611.14355v1.pdf, page 3

[3] arXiv 2611.14355v1.pdf, page 6

[4] arXiv 2611.14355v1.pdf, page 7

[P2] AOP-Smart-Finder: A High-Throughput Human-AI Hybrid Framework for AOP Discovery and Evidence Mapping from Large-Scale Literature

Qinjiang Niu, Lu Yan · arXiv 2611.11349v1 · https://arxiv.org/abs/2611.11349v1 · relevance 0.27

TL;DR: AOP-Smart-Finder is a high-throughput human-AI hybrid system that uses large language models and official AOP-Wiki knowledge to screen tens of thousands of papers, extract thousands of biological events and key-event relationships, and build curated mechanistic networks for toxicity and disease studies.

Problem
The rapid growth of scientific literature and the heterogeneous terminology of biological mechanisms make it hard for researchers to integrate scattered evidence, identify event connections, and construct mechanistic networks [1]. Existing tools such as AOP-helpFinder rely on predefined term matching and co-occurrence statistics, which limit their ability to reveal causal direction, sequence, and detailed mechanisms within single publications [2].

Method
AOP-Smart-Finder combines Large Language Models (LLMs) with a retrieval-augmented generation (RAG) engine (AOP-Smart) that accesses the official AOP-Wiki knowledge base, and adds Python-based automation for data handling [1][2][3]. It offers two analysis modes: (1) single-compound screening that seeks chains like compound -> Event A -> Event B, and (2) event-set-driven screening based on user-provided prompts [3]. The workflow parses titles, abstracts, and DOIs into JSON, uses multithreaded LLM calls (default 20 threads) for initial screening, then extracts upstream/downstream events, species, and literature links; AOP-Smart supplies KE mapping and semantic standardization, after which manual curation and Weight-of-Evidence (WoE) evaluation finalize the network [3][4][5][6].

Key results
- In the DEHP case, the system processed 12,983 articles in 30 min, retained 1,093 for analysis, and extracted 1,589 biological events (458 official KEs and 1,131 candidate KEs) plus 2,519 KERs [4]. A random check of 100 mapped KEs showed 100 % accuracy [4].
- In the ALS case, 437 of 500 selected articles were analyzed, yielding 55 distinct stressors, 278 biological events (85 mapped to official KEs), and 548 event relationships [5]. After manual curation, a mechanistic network with 16 nodes and 23 KERs was built, supported by 108 distinct publications; WoE ratings were High for 13 KERs, Moderate for 10, and none Low [5][6].

Limitations
The paper notes that AOP-Smart-Finder still has limitations, but the excerpt does not detail them beyond a brief statement that "there is still" work to be done [7]. Additionally, the authors acknowledge that LLMs can generate relationships lacking textual evidence or contain factual errors, and that diverse terminology and mapping to standardized entries can affect extraction consistency and accuracy [2][6].

Sources

[1] arXiv 2611.11349v1.pdf, arXiv abstract

[2] arXiv 2611.11349v1.pdf, page 2

[3] arXiv 2611.11349v1.pdf, page 4

[4] arXiv 2611.11349v1.pdf, page 6

[5] arXiv 2611.11349v1.pdf, page 8

[6] arXiv 2611.11349v1.pdf, page 10

[7] arXiv 2611.11349v1.pdf, page 13

[P3] ROAM: Retrieval-Augmented Optimization for Feasible Multi-Day Itinerary Planning

Ye Zihao, Lin Xiaoheng · arXiv 2611.09407v1 · https://arxiv.org/abs/2611.09407v1 · relevance 0.25

TL;DR: ROAM dramatically increases the number of feasible attractions in multi-day itineraries while cutting travel dispersion by using retrieval-augmented optimization [1].

Problem
Large language models can draft plausible travel itineraries but often produce infeasible plans that ignore geography, opening hours, and cannot reliably organize multi-day trips; hard opening-hour constraints and real road-network travel times make spontaneous multi-day itinerary planning challenging [1][2].

Method
ROAM implements a five-stage agentic pipeline: (1) an LLM parses the free-form request into structured constraints such as city, number of days, date range, optional hotel and exclusion set; (2) the request is embedded and a compact candidate set is retrieved from a curated POI database (top-K = 19, total 510 POIs across four Chinese cities) using a Chinese semantic embedding model; (3) candidates are partitioned into day groups via maximal-clique analysis and K-means clustering; (4) each day's order is optimized with a quality-biased simulated-annealing TSP solver and hard opening-hour violations are repaired by a greedy exchange; (5) a spatiotemporal RAG layer injects weather forecasts, real road distances, and POI operating hours into a prompt, and the LLM streams a bracket-balanced JSON itinerary, while an interactive ReAct-style replanning agent reuses stages 2-4 to handle natural-language edits and re-check feasibility [4][5][3][5][3][4][1][3][1][2][4][5].

Key results
- ROAM averages 27.5 POIs per three-day plan, compared with 19.3 for a bare LLM and 17.0 for a retrieval-only baseline, yielding +42 % and +62 % coverage gains respectively [5].
- Route dispersion drops from 2,101 m (baseline) to 677 m, a 3.1× reduction; the bare LLM's dispersion (633 m) is close but its lower coverage explains the difference [5].
- Under rainy forecasts ROAM keeps 26.25 POIs on average; in Fuzhou and Guangzhou coverage rises (28->33 and 21->33) while Shanghai falls to 15 due to sparse indoor POIs [5].
- For sightseeing queries ROAM achieves 34.5 POIs (2.2× bare LLM, 2.0× baseline) and 11.8 % compactness (5.9× bare LLM's 2.0 %) while maintaining route dispersion of 993.3 m [5].
- Ablations show that spatial optimization is the dominant factor for route efficiency and that hard time-window repair prevents roughly 30 % of attractions from violating operating hours [1].

Limitations
Evaluation is offline, measuring only plan metrics without traveler satisfaction studies, and the query suite is limited to 36 queries across four cities [6].
All results come from a single run per query; no variance or statistical significance testing is provided, so reported differences are effect sizes rather than statistically validated gains [6].
The experiments use a single LLM family; cross-model generality remains untested despite a brief spot-check of a second model [6].
Two metrics (semantic match, daily balance) saturate and one (total travel time for the baseline) has a structural artifact, limiting conclusions drawn from those columns [6].
Remaining failure modes include ambiguous requests (no clarifying questions), missing POI categories (generic plans returned), conflicting constraints that force POI drops, and non-trivial cost/latency that grows with candidate set size despite caching [6].

Sources

[1] arXiv 2611.09407v1.pdf, arXiv abstract

[2] arXiv 2611.09407v1.pdf, page 2

[3] arXiv 2611.09407v1.pdf, page 3

[4] arXiv 2611.09407v1.pdf, page 5

[5] arXiv 2611.09407v1.pdf, page 6

[6] arXiv 2611.09407v1.pdf, page 8

[P4] Controlled Knowledge Engineering for Taxonomy Induction with Tool-Grounded LLM Agents and Runtime-Adaptive Policies

Jonas Limniatis, Bennet Schaarmann, Georg Reuber, Robert Stahlbock · arXiv 2611.13966v1 · https://arxiv.org/abs/2611.13966v1 · relevance 0.25

TL;DR: TIGA is a tool-grounded LLM agent framework that improves taxonomy induction's recall and overall F1 by using explicit state, constrained actions, memory, and adaptive policies, but it incurs substantially higher computational cost and does not uniformly beat baselines on precision.

Problem
Automated taxonomy construction must produce hierarchies that are complete, structurally valid, and semantically coherent, yet free-form LLM generation often omits or hallucinates entities, creates invalid relations, and violates consistent organizational principles[1].

Method
The authors propose TIGA (Taxonomy Induction with Tool-Grounded Agents), which treats taxonomy induction as an iterative decision process over an explicit state and a constrained action space of four tools (add_entity, remove_entity, view_subtree, update_policy) that modify the taxonomy (except view_subtree, which is read-only) and log rationales in a memory buffer [3][8].
TIGA incorporates a runtime-adaptive organizational policy that the agent can revise during inference and an optional global reviewer that evaluates the taxonomy against the candidate set and active policy[1].
Experiments use two LLM backbones--a proprietary GPT-5.4-mini2 model and an open-weight Gemma-4-31b model--under identical inputs and temperature 0.0, with five trials per configuration and reporting mean ± standard deviation[4][5].

Key results
Across four benchmark taxonomies, TIGA attains the highest ancestor-level (F1a) or edge-level (F1e) F1 in 7 of 8 evaluated settings[5].
Its largest gains are concentrated in both recall (R_e) and edge-level F1 (F1_e), with improvements of 8.1 percentage points on GProd, 7.9 points on DBLP for recall, and a 7.4-point increase in edge-level F1 on DBLP [5].
Ancestor-level metrics improve modestly but consistently, with 3.8-5.3 % increases in F1a and recalla on

TExEval-2 and GProd[5].
On some precision measures TIGA falls short; the biggest gap is a 13.5 pp lower edge-level precision on DBLP relative to the HF baseline[5].
Ablation shows that removing the policy module reduces F1a by 3.6 % (WordNet) and 2.7 % (TExEval-2) and F1e by 4.1 % and 4.5 % respectively; removing the memory yields drops of 3.1 %-5.8 %; adding the reviewer boosts ancestor-level F1a by 1.2 % and edge-level F1e by 8.0 % on WordNet but slightly harms both metrics on TExEval-2 [6][6].
These quality gains come with substantially higher token consumption, API-call volume, and inference time compared to single-shot and iterative prompting baselines [1].

Limitations
TIGA's higher computational overhead (more model interactions, tokens, and wall-clock time) limits its practicality when resources are constrained[1][7].
It does not uniformly outperform baselines on precision and shows variability across domains and model backbones, indicating that its benefits are not universal[5][7].
The global reviewer's effect is dataset-dependent and can reduce performance when the revision budget is insufficient[6][7].
The study evaluates only four benchmark taxonomies, so generalization to other domains remains unverified[1].

Sources

[1] arXiv 2611.13966v1.pdf, arXiv abstract

[3] arXiv 2611.13966v1.pdf, page 7

[4] arXiv 2611.13966v1.pdf, page 11

[5] arXiv 2611.13966v1.pdf, page 12

[6] arXiv 2611.13966v1.pdf, page 15

[7] arXiv 2611.13966v1.pdf, page 19

[8] arXiv 2611.13966v1.pdf, page 27

[P5] Definitions, Label Names, or Demonstrations? Auditing What Governs LLM Annotation

Viraj Mishra, Shubham Gupta, Dhruv Gupta, Yuehong Cassandra Tai et al. · arXiv 2611.09457v1 · https://arxiv.org/abs/2611.09457v1 · relevance 0.25

TL;DR: The paper audits whether LLMs annotate texts by following supplied codebook definitions or by preserving original label mappings, finding that models usually stick to the original mapping and that retrieval-augmented prompting boosts accuracy without improving definition adherence.

Problem
Large language models are increasingly employed to annotate social-science texts, but high accuracy does not guarantee that models use the researcher-provided category definitions rather than relying on prior label associations or demonstrations [1].

Method
The authors introduce a counterfactual codebook audit that reassigns each definition to a different label (a derangement) and classifies model outputs as matching the reassigned definition, the original label, another valid label, or invalid [2]. They apply this audit before and after task-specific fine-tuning on three political-text datasets (CCC, BFRS, Manifesto) and evaluate retrieval-augmented annotation using Global Example-RAG (retrieving the k nearest training examples as a shared demonstration block) versus Class-wise retrieval and direct k-nearest-neighbor (kNN) prediction [2][4].

Key results
- Under definition reassignment, original-label agreement exceeds definition adherence on all datasets: 55.80 % vs 10.75 % on CCC, 20.93 % vs 12.87 % on BFRS, and 30.25 % vs 3.24 % on Manifesto [3]. Fine-tuning changes vary; for Mistral-7B on CCC original-label agreement increases by +10.81 pp while definition adherence decreases by 3.92 pp, and for Llama-3.1-8B definition adherence increases by +1.15 pp on BFRS and +1.96 pp on Manifesto [3]. Similar patterns are reported for Llama-3.1-8B across datasets [3].
- Retrieval improves weighted F1: adding Global demonstrations to Mistral's codebook prompt raises development/test F1 on Manifesto from .416/.404 to .580/.551 and on BFRS from .545/.548 to .794/.774 [4].
- Global demo-only prompting outperforms codebook-only in 11 of 12 paired bootstrap comparisons and exceeds label-only prompting numerically in all 12 cases [4].

Limitations
The study is limited to three English-language political-text datasets and a set of open-weight models; other models may balance definitions, labels, and demonstrations differently [6]. The reassignment audit preserves category meanings, so it tests responsiveness to a supplied label-definition mapping rather than learning new category boundaries [6]. The audit reports observable agreement but does not reveal internal causal mechanisms, and the conflict tests hold queries, retrieved texts, ordering, decoding, and parsing fixed [6].

Sources

[1] arXiv 2611.09457v1.pdf, arXiv abstract

[2] arXiv 2611.09457v1.pdf, page 3

[3] arXiv 2611.09457v1.pdf, page 5

[4] arXiv 2611.09457v1.pdf, page 6

[6] arXiv 2611.09457v1.pdf, page 8